

Below is a **full implementation blueprint** for:

MSME Copilot Twin

AI Retrofit Kit for Machine Intelligence and Worker Assistance

This idea is very suitable for MSME Idea Hackathon 6.0 because the hackathon expects ideas that lead to a **working prototype, model, proof-of-concept, or product**. It also directly matches the Industry 4.0/5.0 theme, which encourages **IoT, digital twins, AI, machine vision, cyber-physical systems, affordable sensors, machinery retrofitting, predictive maintenance tools, digital work instructions, production monitoring, and worker-assistance systems**.

1. Product Vision

Product Name

MSME Copilot Twin: AI Retrofit Kit for Machine Intelligence and Worker Assistance

One-line Product Pitch

MSME Copilot Twin is an AI-powered retrofit kit that converts old MSME machines into intelligent, monitored, and worker-assisted systems using sensor-based digital twins and a private multilingual factory copilot.

Target Users

This product is not only for students or colleges. It is for:

- Small machining workshops
- Textile units
- Packaging units
- Fabrication shops
- Plastic moulding units

- Food-processing units
 - Pump/compressor-based MSMEs
 - Training centres and ITIs
 - Maintenance teams in small factories
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2. Core Problem

Most MSMEs use machines that are mechanically strong but digitally blind.

Typical problems:

- Old machines have no PLC or monitoring system.
- Owners do not know actual machine utilization.
- Maintenance is reactive, not predictive.
- Operators depend on senior technicians for troubleshooting.
- Manuals and SOPs are not used effectively.
- Energy wastage is not measured machine-wise.
- Small factories cannot afford full Industry 4.0 solutions.

So the product must solve one clear problem:

How can an old MSME machine become intelligent without replacing it?

3. Final Product Concept

The product has two connected parts:

Part A: Retrofit Digital Twin

This converts a physical machine into a live digital machine model using sensors.

Part B: FactoryGPT-Safe Copilot

This gives workers and maintenance teams a private AI assistant trained on machine manuals, SOPs, alerts, sensor data, and maintenance history.

4. Complete System Architecture

Old MSME Machine



Sensor Layer

(Current + Vibration + Sound + Temperature + Optional RPM)



Edge Data Acquisition

ESP32 / STM32 / Raspberry Pi



Edge AI Machine-State Detection

Idle / Running / Overload / Abnormal / Maintenance Needed



Digital Twin Dashboard

Utilization + Health Score + Energy + Alerts



Private Factory Copilot

Manual Q&A + SOP + Safety + Maintenance Checklist



Worker Guidance

Gujarati / Hindi / English

5. Recommended MVP Machine

For the first prototype, do **not** try to cover all machines.

Choose one of these:

Best MVP Option

Single-phase motor-based drilling/lathe/compressor test setup

Why this is best:

- Easy to attach current sensor
 - Easy to create normal and abnormal operating conditions
 - Easy to demonstrate vibration, heating, overload, idle running
 - Looks industrial enough for hackathon
 - Can later be tested in a real MSME workshop
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6. Hardware Implementation

6.1 Sensor Layer

Parameter	Sensor Option	Purpose
Current	SCT-013 current clamp / ACS758	Machine load, idle/running detection, overload
Vibration	ADXL345 / MPU6050 / industrial accelerometer	Bearing issue, imbalance, abnormal vibration
Sound	INMP441 I2S microphone / industrial acoustic sensor	Abnormal machine sound, cutting/chatter sound
Temperature	DS18B20 / MLX90614 / thermocouple	Motor heating, bearing heating, unsafe operation
RPM Optional	Hall sensor / optical encoder	Speed variation, belt slip, load variation
Power Optional	PZEM-004T / industrial energy meter module	kWh, voltage, power factor approximation

For a hackathon prototype, use:

Current sensor + vibration sensor + temperature sensor + microphone

That is enough to demonstrate machine intelligence.

6.2 Edge Device Options

Low-cost MVP

ESP32 + Raspberry Pi

ESP32 collects sensor data. Raspberry Pi performs AI inference, dashboard hosting, and copilot backend.

Stronger Product Prototype

STM32 / ESP32-S3 + Raspberry Pi 5 / Jetson Orin Nano

Use this if you want better edge AI, computer vision, or local SLM support.

6.3 Hardware Block Diagram

Machine Motor / Machine Body



Current Clamp Sensor —┐

Vibration Sensor —┐

Temperature Sensor —┐

Microphone Sensor —┐

RPM Sensor Optional —┐



ESP32 / STM32 Data Acquisition Unit

↓ MQTT / Serial / Wi-Fi

Raspberry Pi / Edge Gateway



AI Inference + Database + API



Dashboard + Factory Copilot

7. Software Architecture

7.1 Main Software Modules

Module	Function
Sensor firmware	Reads current, vibration, temperature, sound
Edge ingestion service	Receives sensor data using MQTT/HTTP
Feature extraction engine	Converts raw signal into useful features
AI inference engine	Detects machine state and abnormal condition
Digital twin database	Stores machine profile, readings, events
Dashboard	Shows live machine health and utilization
RAG knowledge base	Stores manuals, SOPs, safety documents
Factory copilot	Answers worker questions safely
Agent workflow engine	Creates checklist, alerts, reports
Admin panel	Adds machines, documents, users, cost settings

7.2 Recommended Tech Stack

Layer	Technology
Firmware	Arduino C / ESP-IDF / MicroPython
Messaging	MQTT
Backend	FastAPI
Database	PostgreSQL + TimescaleDB or SQLite for MVP

Layer	Technology
Time-series dashboard	Grafana or custom React dashboard
AI/ML	Python, scikit-learn, PyTorch, ONNX Runtime
RAG	FAISS / ChromaDB
Document processing	PDF parser, OCR if needed
SLM/LLM	Local small model or private API during MVP
Frontend	React / Next.js
Deployment	Docker
Edge device	Raspberry Pi / Jetson

8. Digital Twin Design

Do not make “digital twin” only a 3D animation. In industry, your digital twin should be a **live data model of the machine**.

8.1 Digital Twin Data Model

Each machine should have:

- Machine ID
- Machine type
- Rated power
- Motor rating
- Installed location
- Sensor configuration
- Operating states
- Utilization history
- Energy history
- Alert history
- Maintenance history
- Manuals and SOPs
- Health score

8.2 Machine States

The AI should classify the machine into states:

OFF
IDLE
RUNNING_NORMAL
RUNNING_LOADED
OVERLOAD
ABNORMAL_VIBRATION
ABNORMAL_SOUND
OVERHEATING
MAINTENANCE_REQUIRED

8.3 Digital Twin Dashboard

Dashboard should show:

- Live machine state
- Current consumption
- Vibration level
- Temperature
- Sound abnormality score
- Runtime today
- Idle time today
- Energy estimate
- Health score
- Alerts
- Suggested action
- Maintenance checklist

Example dashboard output:

Machine: Drill Press 01
Current State: Running Loaded
Health Score: 78/100

Idle Time Today: 1 hour 20 minutes
Abnormal Vibration: Moderate
Temperature: 54°C
Suggested Action: Check bearing alignment within 48 hours

9. AI/ML Implementation

9.1 Data Collection Plan

Collect data in five machine conditions:

Condition	Example
OFF	Machine disconnected
IDLE	Motor running without load
NORMAL LOAD	Normal drilling/cutting/compressor load
OVERLOAD	Extra load, tight belt, heavy cutting
ABNORMAL	Artificial vibration, loose mounting, heating, unusual sound

For each condition, collect:

- Current data
 - Vibration data
 - Sound data
 - Temperature data
 - Timestamp
 - Machine state label
 - Operator note
-

9.2 Sampling Recommendation

Sensor	Suggested Sampling
Current	500 Hz to 2 kHz

Sensor	Suggested Sampling
Vibration	1 kHz to 5 kHz
Sound	8 kHz to 16 kHz
Temperature	1 sample/second
RPM	1 to 10 samples/second

For MVP, even lower sampling can work if the demonstration is focused on state detection.

9.3 Feature Extraction

From current signal:

- RMS current
- Peak current
- Current variation
- Load factor
- On/off pattern
- Idle power estimate

From vibration:

- RMS vibration
- Peak-to-peak value
- Kurtosis
- Crest factor
- Dominant frequency
- Band power

From sound:

- Sound energy

- Zero-crossing rate
- Spectral centroid
- MFCC features
- Abnormal acoustic score

From temperature:

- Current temperature
 - Rate of rise
 - Overheat threshold
 - Cooling pattern
-

9.4 AI Model Strategy

Stage 1: Rule + Threshold Model

Use this first because it is explainable and easy to debug.

Example:

If current < threshold and vibration low → OFF

If current high and vibration normal → RUNNING_NORMAL

If current high and vibration high → ABNORMAL_VIBRATION

If temperature rising continuously → OVERHEATING_RISK

Stage 2: Classical ML Model

Use after collecting labelled data.

Good models:

- Random Forest
- XGBoost
- Isolation Forest
- One-Class SVM
- Logistic Regression for baseline

Stage 3: Deep Learning Model

Use only after enough data.

Possible models:

- 1D CNN for sensor signals
- LSTM/GRU for time-series
- Autoencoder for anomaly detection

For hackathon, a **hybrid rule + ML model** is the safest and most explainable.

10. Health Score Calculation

Create a simple machine health score.

Example:

Health Score = 100

- Vibration Penalty
- Temperature Penalty
- Overload Penalty
- Idle Penalty
- Abnormal Sound Penalty

Example output:

Machine Health Score: 72/100

Reasons:

- Vibration level increased by 38%
- Motor temperature crossed safe threshold twice
- Idle running detected for 45 minutes

This makes the product understandable for MSME owners.

11. Energy Intelligence Module

The product should not only show current; it should estimate business loss.

11.1 Energy Metrics

Show:

- Runtime energy
- Idle energy
- Estimated monthly wastage
- Overload energy loss
- Cost per machine-hour
- Cost per job cycle

Example:

Idle Running Today: 1.5 hours

Estimated Energy Wasted: 2.1 kWh

Approximate Cost Lost: ₹18.90

Monthly Projection: ₹567

The hackathon document values ideas that are scalable, reduce cost, improve quality of life, and focus on clean/green energy initiatives . So cost-saving and energy-saving reports should be visible in the demo.

12. FactoryGPT-Safe Copilot

This is the second major innovation.

12.1 Purpose

The copilot should help workers and maintenance teams by answering questions such as:

- Why is this alert coming?
- What should I check first?
- What is the safe shutdown procedure?
- How do I perform daily maintenance?

- What does this machine error code mean?
 - How can I reduce idle energy?
 - What is the SOP for belt replacement?
 - Translate this maintenance step into Gujarati.
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12.2 Knowledge Sources

Upload:

Machine manual

Maintenance SOP

Safety SOP

Troubleshooting guide

Previous maintenance logs

Sensor alert history

Energy reports

Operator notes

The copilot should answer using only these approved sources.

12.3 RAG Pipeline

PDF / SOP / Manual Upload

↓

Text Extraction

↓

Chunking

↓

Embedding Generation

↓

Vector Database

↓

User Question

↓

Relevant Chunk Retrieval

↓

SLM Answer Generation

↓

Safety Check



Answer with Source + Confidence

12.4 Copilot Tool Calling

The copilot should not only chat. It should call system tools.

Example tools:

```
get_machine_status(machine_id)
get_recent_alerts(machine_id)
get_energy_report(machine_id)
search_manual(query)
generate_maintenance_checklist(machine_id)
create_service_ticket(machine_id, issue)
calculate_idle_cost(machine_id)
```

Example user query:

Why is Machine 1 giving an abnormal vibration alert?

Copilot response should combine sensor data and manual guidance:

Machine 1 has shown vibration above normal range for the last 18 minutes.
The current load is normal, but vibration increased by 42%.

Possible causes:

1. Loose mounting
2. Bearing wear
3. Tool imbalance

Recommended safe checklist:

1. Stop the machine using normal shutdown.
 2. Wait until moving parts stop completely.
 3. Inspect mounting bolts.
 4. Check bearing temperature.
 5. Record observation in maintenance log.
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13. Safety Design

This part is very important. For industrial products, the AI should not directly control hazardous machinery.

Safety Rules

The copilot must never say:

- Bypass safety guard
- Ignore vibration
- Continue operating during overheating
- Disable emergency stop
- Open electrical panel without authorization
- Repair live electrical components

Recommended Safety Architecture

AI Copilot



Advisory Guidance Only



Human Confirmation



Supervisor Approval



Maintenance Action

For the first prototype, keep the product **read-only**.

Do not connect automatic machine shutdown unless supervised by an industrial safety expert.

14. Dashboard Pages

Page 1: Machine Overview

Shows all machines:

Machine 1: Healthy
Machine 2: Abnormal vibration
Machine 3: Idle for 2 hours
Machine 4: Overheating risk

Page 2: Machine Digital Twin

Shows one machine in detail:

- Current state
- Live sensor graphs
- Health score
- Utilization
- Energy cost
- Alerts
- Maintenance history

Page 3: Copilot

Chat interface:

Ask:

“Why is Machine 2 showing vibration alert?”

“Generate maintenance checklist.”

“How much energy was wasted in idle mode?”

Page 4: Reports

Generate:

- Daily machine report
- Energy saving report
- Maintenance report
- Alert summary
- Operator checklist
- Monthly MSME owner report

15. Database Design

Machine Table

machine_id
machine_name
machine_type
rated_power
location
installation_date
sensor_config

Sensor Reading Table

reading_id
machine_id
timestamp
current_value
vibration_x
vibration_y
vibration_z
sound_level
temperature
rpm

Feature Table

feature_id
machine_id
timestamp
rms_current
rms_vibration
sound_energy
temperature_rate
dominant_frequency

Event Table

event_id
machine_id
timestamp
event_type

severity
model_confidence
recommended_action

Document Table

document_id
machine_id
document_type
file_name
uploaded_by
upload_date

Maintenance Log Table

log_id
machine_id
issue
action_taken
technician
date
remarks

16. MVP Development Plan

Phase 1: Problem Finalization

Duration: 3–5 days

Tasks:

- Select one target machine.
- Define 3–5 operating states.
- Decide sensors.
- Prepare product pitch.
- Prepare concept note.
- Prepare block diagram without identity-revealing information.

The hackathon instruction says not to reveal phone, email, brand name, or incubatee/MSME identity in the block diagram and concept note, so keep the proposal anonymized.

Phase 2: Hardware Prototype

Duration: 10–15 days

Build:

- Sensor mounting
- ESP32 data reading
- Raspberry Pi gateway
- MQTT communication
- Local database logging
- Basic live graph

Deliverable:

Machine sensor data visible on dashboard

Phase 3: Machine State Detection

Duration: 15–20 days

Tasks:

- Collect data in OFF, IDLE, NORMAL, OVERLOAD, ABNORMAL states.
- Extract features.
- Train simple classifier.
- Display machine state live.
- Generate alerts.

Deliverable:

Machine state detected automatically

Phase 4: Digital Twin Dashboard

Duration: 10–15 days

Tasks:

- Create dashboard.
- Add health score.
- Add utilization.
- Add energy cost.
- Add alert history.

Deliverable:

Digital twin dashboard for one machine

Phase 5: FactoryGPT-Safe Copilot

Duration: 15–20 days

Tasks:

- Upload machine manual/SOP.
- Build RAG pipeline.
- Add copilot interface.
- Connect copilot to machine status API.
- Add safety rules.
- Add checklist generation.

Deliverable:

Worker can ask machine-related questions and receive safe, grounded answers

Phase 6: Field Validation

Duration: 15–30 days

Tasks:

- Test on one real MSME machine.
- Record feedback from operator/owner.
- Compare AI state detection with human observation.
- Calculate idle-time and energy-saving estimate.
- Improve dashboard and copilot answers.

Deliverable:

Field-tested prototype report

17. 90-Day Implementation Roadmap

Week	Work
1	Finalize machine, use case, sensors, product scope
2	Build sensor acquisition hardware
3	Store live data and show basic dashboard
4	Collect machine-state data
5	Train first ML model
6	Build digital twin dashboard
7	Add energy and utilization analytics
8	Build document upload and RAG system
9	Add FactoryGPT-Safe copilot
10	Add tool calling and checklist

Week	Work
	generation
11	Test in real/simulated MSME environment
12	Prepare demo video, report, budget, pitch deck

18. Evaluation Metrics

Technical Metrics

Metric	Target
Machine state detection accuracy	85–90% for MVP
Alert latency	Less than 5 seconds
Sensor data loss	Less than 5%
Dashboard refresh	1–5 seconds
Idle detection accuracy	More than 85%
RAG answer grounding	More than 90% answers with source
Unsafe answer rate	0 critical unsafe answers

Business Metrics

Metric	Target
Idle energy reduction	10–20% potential saving
Maintenance response time	Reduced by 20–30%
Breakdown early warning	Detect before severe failure
Training time for worker	Reduced using copilot
Manual troubleshooting dependency	Reduced

19. Estimated Budget for ₹15 Lakh Proposal

The hackathon support can go up to ₹15 lakhs for development of idea into prototype . A realistic budget can be:

Head	Approx. Amount
Sensors and industrial-grade DAQ	₹1.50 lakh
Edge devices: Raspberry Pi/Jetson/industrial gateway	₹1.50 lakh
Enclosure, PCB, wiring, connectors, mounting	₹1.00 lakh
Test machine setup and instrumentation	₹1.50 lakh
Software development and dashboard	₹2.00 lakh
AI/ML model development and testing	₹2.00 lakh
RAG/copilot development	₹1.50 lakh
Field testing in MSME unit	₹1.00 lakh
Cloud/local server, storage, security	₹0.75 lakh
Documentation, demo video, UI/UX, reports	₹0.75 lakh
IP/prototype refinement/certification preparation	₹1.00 lakh
Contingency	₹0.50 lakh
Total	₹15.00 lakh

For actual student MVP, you can start with ₹25,000–₹60,000 using low-cost sensors and Raspberry Pi.

20. Team Structure

A 5-member student team is ideal.

Role	Responsibility
Embedded/IoT Lead	Sensors, ESP32, data acquisition
AI/ML Lead	Feature extraction, anomaly detection, model evaluation
GenAI/RAG Lead	FactoryGPT-Safe, document Q&A, safety guardrails
Backend/Cloud Lead	APIs, database, MQTT, deployment
Frontend/Product Lead	Dashboard, reports, demo, user feedback

Faculty mentor role:

- Product direction
 - MSME contact
 - Safety review
 - Proposal review
 - Evaluation methodology
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21. What Makes This Novel

Do not claim that each individual part is completely new. Sensors, dashboards, and chatbots already exist.

The novelty should be framed like this:

Novelty Statement

The novelty of MSME Copilot Twin lies in combining sensor-based retrofit digital twins for old MSME machines with a private multilingual factory copilot that uses live machine data, manuals, SOPs, and maintenance history

to provide safe, contextual, worker-friendly guidance. Unlike conventional IoT monitoring tools, it focuses on affordable retrofitting, machine-state intelligence, energy insight, maintenance assistance, and human-centric Industry 5.0 adoption for MSMEs.

This is a credible novelty statement.

22. Prototype Demonstration Scenario

Use this demo flow:

Step 1: Normal Running

Show machine current, vibration, temperature.

Dashboard:

State: RUNNING_NORMAL

Health: 92/100

Step 2: Idle Running

Run motor without useful load.

Dashboard:

State: IDLE

Energy Wasted: ₹12 estimated today

Advice: Switch off during non-production period

Step 3: Abnormal Vibration

Create controlled imbalance or loose mount.

Dashboard:

State: ABNORMAL_VIBRATION

Severity: Medium

Suggested Action: Inspect mounting and bearing

Step 4: Ask Copilot

Question:

Why is vibration alert coming?

Copilot:

The vibration level is above the normal baseline for this machine.
Possible causes are loose mounting, bearing wear, or imbalance.
Follow safe shutdown and inspection checklist.

Step 5: Generate Report

Output:

Daily Machine Intelligence Report

- Runtime: 5.2 hours
- Idle time: 1.1 hours
- Abnormal events: 2
- Estimated energy wastage: ₹34
- Maintenance recommendation: Bearing inspection

This will look like a real product, not a student demo.

23. Risk and Mitigation

Risk	Mitigation
Sensor noise	Use filtering, calibration, averaging
Low data availability	Start with rule-based model and gradually collect data
Wrong AI answer	Use RAG with source citation and safety guardrails
Industrial safety issue	Keep system advisory/read-only in MVP
Too broad scope	Start with one machine and 3–5 states
Weak novelty perception	Emphasize retrofit digital twin + worker copilot + MSME affordability
Hallucination	Force answer from uploaded

Risk	Mitigation
	manual/SOP and show confidence
Hardware failure	Use modular sensor design and watchdog logging

24. Minimum Demo Version

If time is short, build this minimum version:

One motor-based machine

- + current sensor
- + vibration sensor
- + temperature sensor
- + Raspberry Pi dashboard
- + machine state detection
- + manual/SOP chatbot
- + maintenance checklist generation

This is enough for a strong PoC.

25. Final Submission Positioning

Do not write the title as:

IoT-Based Machine Monitoring System

That sounds common.

Use this:

MSME Copilot Twin: AI Retrofit Kit for Machine Intelligence and Worker Assistance

Final Abstract

MSME Copilot Twin is an AI-powered retrofit solution designed to modernize old MSME machines without replacing existing infrastructure. The system

uses current, vibration, sound, and temperature sensors to create a live digital twin of the machine, enabling machine-state detection, utilization monitoring, abnormal-condition alerts, and energy-loss estimation. In addition, a private multilingual FactoryGPT-Safe copilot uses machine manuals, SOPs, maintenance logs, and live sensor alerts to guide workers through safe troubleshooting, maintenance checklists, and energy-saving actions. The product supports Industry 4.0 and Industry 5.0 adoption by combining affordable retrofitting, edge AI, digital twin intelligence, and human-centric worker assistance for MSMEs.

This is a strong, product-level, hackathon-ready concept with real implementation depth.